The logical construction of geometry

Why a geometry course begins with words it refuses to define, and how everything else is built on top of them.

LESSON 1 | 1 × 40 MIN | GEOMETRY 10, §1 | IGCSE E4.1

LESSON CLOCK

7' | 11' | 11' | 7' | 4'

The three undefined terms	7 min
Axiom, definition, theorem	11 min
How a proof is built	11 min
Converses	7 min
Homework	4 min

LEARNING OBJECTIVES

- Name the undefined terms of geometry and say why they cannot be defined.
- Distinguish an axiom, a definition and a theorem.
- Describe the structure of a deductive proof.
- Give an example of a statement whose converse is false.

TEXTBOOK

National	pp. 3–8
Cambridge	Core & Extended, pp. 246–248
Workbook	Discussion sheet L1

01 Explanation

Three words we refuse to define

Every definition explains a new word using older words. Follow the chain back far enough and it must stop somewhere, or it runs in a circle. Geometry stops at three:

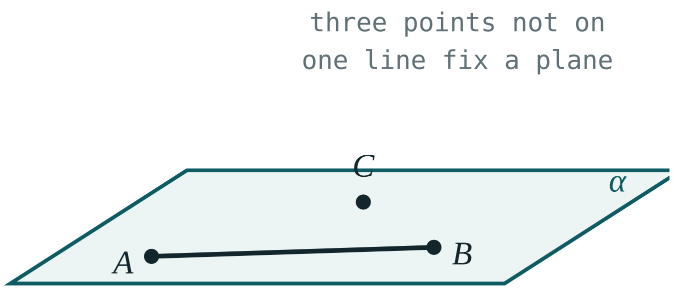
THE UNDEFINED TERMS

point · **line** · **plane**. They are not explained. They are described by how they behave, and that behaviour is set out by the axioms.

“A point is that which has no part” — Euclid's own attempt — defines “point” using “part”, which he never defines either. Modern geometry is honest about the stopping place.

Three kinds of statement

KIND	STATUS	EXAMPLE
axiom	accepted without proof	Through any two points passes exactly one line
definition	a name given to a described object	A trapezium is a quadrilateral with exactly one pair of parallel sides
theorem	proved from axioms, definitions and earlier theorems	The angles of a triangle sum to $180°$



Three axioms in one picture: two points fix a line, three non-collinear points fix a plane, a line with two points in a plane lies in it.

A DEFINITION IS NEVER TRUE OR FALSE

It is a naming convention. Arguing whether “a square is a rectangle” is *true* is arguing about a definition — and the answer is simply whichever the textbook chose. In this course a square **is** a rectangle.

The shape of a proof, and the converse trap

A proof is a chain: **Given** → statements, each justified by an axiom, a definition or an earlier theorem → **therefore** the conclusion. Every line needs a reason written beside it.

The **converse** swaps the given and the conclusion. It is a different statement and needs its own proof — sometimes it is false:

STATEMENT	CONVERSE	CONVERSE TRUE?
A square has four equal sides	A shape with four equal sides is a square	no — a rhombus
Vertical angles are equal	Equal angles are vertical	no
$a^2 + b^2 = c^2 \Rightarrow$ right angle	right angle $\Rightarrow a^2 + b^2 = c^2$	yes — both are theorems

02 Worked examples

EXAMPLE 1 **Classify: “Through a point not on a line there is exactly one parallel to it.”**

1. Can it be proved from the others?

   For two thousand years people tried and failed.

2. It is assumed.

   Euclid’s fifth postulate.

3. Denying it gives a different, consistent geometry.

ANSWER

An axiom

EXAMPLE 2 **Write the converse of “If a quadrilateral is a parallelogram then its diagonals bisect each other”, and say whether it is true.**

1. Swap given and conclusion.
   If the diagonals bisect each other then it is a parallelogram.

2. Is it provable?
   Yes — a standard Grade 8 theorem.

ANSWER

True; it is the standard test for a parallelogram

EXAMPLE 3 **Why can “line” not be defined as “the shortest path between two points”?**

1. “Shortest” needs distance.
   Distance is defined using lines.

2. The definition is circular.

ANSWER

It defines a line in terms of something already defined by lines

03 Interactive model

Ask the class for a definition of “point”, then keep asking what each new word means.

Axiom, definition or theorem?

"Through two points passes exactly one line."

- axiom
- definition
- theorem
- corollary

"A rhombus is a parallelogram with four equal sides."

- axiom
- definition
- theorem
- converse

"The angles of a triangle sum to $180°$."

- axiom
- definition
- theorem
- undefined term

The undefined terms of geometry are:

- point, line, plane
- point, angle, area
- line, circle, square
- length, angle, area

The converse of a true theorem is:

- always true
- always false
- sometimes true
- meaningless

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Axiom (postulate)	Aksioma	Аксиома
Undefined term	Ta'riflanmaydigan tushuncha	Неопределяемое понятие
Definition	Ta'rif	Определение
Theorem	Teorema	Теорема
Proof	Isbot	Доказательство
Converse	Teskari teorema	Обратная теорема
Corollary	Natija	Следствие
Deduction	Deduksiya	Дедукция
Consistent system	Zid bo'lmagan sistema	Непротиворечивая система
Euclidean geometry	Evklid geometriyasi	Евклидова геометрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

An axiom is:

- proved
- assumed
- a name
- a guess

A theorem must have:

- a diagram
- a proof
- a converse
- a number

Which is **not** undefined in geometry?

- point
- line
- plane
- triangle

Swapping the given and the conclusion gives the:

- corollary
- converse
- axiom
- proof

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Name the three undefined terms

   ANSWER point, line, plane

2. Is “through two points passes one line” an axiom or theorem?

   ANSWER axiom

3. Is “the angles of a triangle sum to 180°” an axiom or theorem?

   ANSWER theorem

4. What is a corollary?

   ANSWER A result following immediately from a theorem

5. Give the converse of “if it is a square it has four right angles”

   ANSWER If it has four right angles it is a square

6. Is that converse true?

   ANSWER no — a rectangle

7. Can a definition be false?

   ANSWER no — it is a naming convention

Medium · the standard question

7 PROBLEMS

1. Classify “vertically opposite angles are equal”

   ANSWER theorem

2. Classify “a trapezium has one pair of parallel sides”

   ANSWER definition

3. Write the converse of Pythagoras and say whether it is true

   ANSWER If $a^2+b^2=c^2$ the angle is right — true

4. Why must some terms be undefined?

   ANSWER To stop the chain of definitions running in a circle

5. Give a false converse of your own

   ANSWER e.g. “equal areas ⇒ congruent”

6. What does “consistent” mean of an axiom system?

   ANSWER No contradiction can be derived from it

7. Name a geometry where the parallel axiom fails

   ANSWER Spherical or hyperbolic geometry

Hard · stretch and reason

7 PROBLEMS

1. Explain why the parallel postulate cannot be proved from the others

   ANSWER Consistent geometries exist in which it is false

2. Give a statement, its converse, and show exactly one is true

   ANSWER e.g. “square ⇒ rhombus” true; converse false

3. Is “a square is a rectangle” true?

   ANSWER Yes under our definition — a naming choice

4. Prove: if two lines meet, they meet in exactly one point

   ANSWER Two common points would force one line, by the first axiom

5. What would happen if we assumed two parallels through a point?

   ANSWER Hyperbolic geometry — consistent, but not Euclidean

6. Distinguish “proof by contradiction” from a direct proof

   ANSWER It assumes the negation and derives an impossibility

7. Why is a diagram not a proof?

   ANSWER It shows one case; a proof must cover all

07 Homework

Homework — 4 tasks

Task 4 should be no more than five lines.

1. Classify five statements from Chapter I as axiom, definition or theorem.
2. Write the converse of three theorems you know and say which are true.
3. Explain, in your own words, why “point” is not defined.
4. Find one statement in everyday life whose converse is false, and write both.

Geometric problems and the methods of solving them

Four standard methods, and a way of choosing between them before you have drawn anything.

LESSON 2 | 1 × 40 MIN | GEOMETRY 10, §2 | IGCSE E4.2

LESSON CLOCK

5' | 12' | 15' | 5' | 3'

Four methods	5 min
Choosing between them	12 min
One problem, three methods	15 min
Setting out	5 min
Homework	3 min

LEARNING OBJECTIVES

- Name the synthetic, coordinate, vector and trigonometric methods.
- Choose a method from the wording of a problem.
- Draw a clear labelled diagram before starting.
- Present a solution with reasons at every line.

TEXTBOOK

National	pp. 9–14
Cambridge	Core & Extended, pp. 249–256
Workbook	Method sheet L2

01 Explanation

Four methods

METHOD	TOOLS	REACH FOR IT WHEN
synthetic	congruence, similarity, circle theorems	the problem is about equal or parallel things
coordinate	axes, distance, gradient, equations	perpendicularity, midpoints, or a right angle is convenient
vector	position vectors, scalar product	collinearity, ratios along a line, angles in space
trigonometric	sine and cosine rules, areas	lengths and angles are mixed in the same problem

THE CHOOSING RULE

Look at what the problem **gives**, not at what it asks. Given angles and lengths — trigonometry. Given a rectangle or a right angle — coordinates. Given ratios along segments — vectors. Given equal sides or parallels — synthetic.

One problem, three ways

Problem. $ABCD$ is a square. M is the midpoint of BC. Prove AM and DM are not perpendicular, and find the angle $\angle AMD$.

METHOD	FIRST LINE
coordinate	$A(0,0), B(2,0), C(2,2), D(0,2), M(2,1)$
vector	$AM = 2i + j, DM = 2i - j$
trigonometric	$AM = DM = \sqrt{5}$, $AD = 2$, then the cosine rule

$$\cos \angle AMD = \frac{5 + 5 - 4}{2 \cdot 5} = 0.6 \Rightarrow \angle AMD \approx 53.1°$$

All three agree. The coordinate method is shortest here because the square hands you the axes.

Setting out

FIVE HEADINGS, EVERY TIME

Given · **Required** · **Diagram** · **Solution**, with a reason on every line · **Answer**, in a sentence with units.

DRAW IT BIG

A diagram smaller than a matchbox hides the very relationship you are looking for. Half a page, labelled, with the given data marked on it — that is where most solutions are actually found.

02 Worked examples

EXAMPLE 1 **Which method: "In triangle ABC, $a = 7$, $b = 9$, $\angle C = 40°$. Find c."**

1. Two sides and the included angle.
   Lengths mixed with an angle.

2. Cosine rule.

ANSWER

Trigonometric method

EXAMPLE 2 **Which method: "Prove the diagonals of a rhombus are perpendicular."**

1. Equal sides given.
   Congruent triangles available.

2. Coordinates also work, but need setting up.

ANSWER

Synthetic (coordinates a close second)

EXAMPLE 3 **Which method: “P divides AB in the ratio $2 : 3$. Show P, Q, R are collinear.”**

1. Ratios along segments, collinearity.
2. Position vectors handle both directly.

ANSWER

Vector method

03 Interactive model

Solve the square problem on the board by two methods and let the class time each.

Pick the method

Two sides and the included angle, find the third side:

- synthetic
- coordinate
- vector
- trigonometric

Prove the midpoints of a quadrilateral form a parallelogram:

- vector
- trigonometric
- none
- measurement

Find the perpendicular distance from a point to a line:

- synthetic
- coordinate
- guessing
- drawing

Prove base angles of an isosceles triangle are equal:

- synthetic
- coordinate
- vector
- trigonometric

The first thing to do with any geometry problem:

- write the formula
- draw a large labelled diagram
- guess the answer
- measure

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Synthetic method	Sintetik usul	Синтетический метод
Coordinate method	Koordinatalar usuli	Координатный метод
Vector method	Vektor usuli	Векторный метод
Trigonometric method	Trigonometrik usul	Тригонометрический метод
Auxiliary line	Yordamchi chiziq	Вспомогательная линия
Construction	Yasash	Построение
Analysis of a problem	Masalani tahlil qilish	Анализ задачи
Given and required	Berilgan va topilishi kerak	Дано и найти
Diagram	Chizma	Чертёж

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The coordinate method is best when the figure has:

- equal sides
- right angles
- circles only
- no data

Collinearity is most directly proved with:

- vectors
- the sine rule
- congruence
- measurement

A reason must be written:

- at the end
- on every line
- only in proofs
- never

An auxiliary line is:

- given
- added by you to help
- always wrong
- a mistake

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the four methods

   ANSWER synthetic, coordinate, vector, trigonometric

2. Best method for the cosine rule situation

   ANSWER trigonometric

3. Best method for collinearity

   ANSWER vector

4. Best method for “prove the diagonals bisect”

   ANSWER synthetic or vector

5. What are the five headings of a solution?

   ANSWER given, required, diagram, solution, answer

6. How large should a diagram be?

   ANSWER about half a page

7. What is an auxiliary line?

   ANSWER A line you add to create a usable figure

Medium · the standard question

7 PROBLEMS

1. Square side 2, M midpoint of BC. Find AM

   ANSWER $\sqrt{5}$

2. Same square; find $\angle AMD$

   ANSWER $\approx 53.1°$

3. Choose a method: find the angle between two space diagonals of a cube

   ANSWER vector

4. Choose a method: prove a midline is half the base

   ANSWER vector or synthetic

5. Choose a method: a triangle with sides 5, 6, 7 — find its largest angle

   ANSWER trigonometric

6. Set up coordinates for a rectangle 6×4

   ANSWER $(0,0), (6,0), (6,4), (0,4)$

7. Why is a big diagram worth the time?

   ANSWER It reveals the relationship you must use

Hard · stretch and reason

7 PROBLEMS

1. Solve the square-and-midpoint problem by all three methods and compare lengths of solution

   ANSWER Coordinate is shortest

2. Prove the medians of a triangle are concurrent, by vectors

   ANSWER Show $\frac{a+b+c}{3}$ lies on each median

3. Prove the midpoints of any quadrilateral form a parallelogram

   ANSWER Each midline is $\frac{1}{2}$ of a diagonal

4. A cube of side 1: find the angle between a face diagonal and a space diagonal

   ANSWER $\approx 35.26°$

5. When would the coordinate method be a bad choice?

   ANSWER When the figure has no right angles or convenient symmetry

6. Give a problem where two methods take about the same time

   ANSWER Diagonals of a rhombus

7. Explain why the vector method extends to space but the plane coordinate method must be extended first

   ANSWER Vectors already carry three components

07 Homework

Homework — 4 tasks

Task 2 needs the diagram, not only the answer.

1. For five problems from the exercise, write only the method you would choose and one sentence of why.
2. Solve the square-and-midpoint problem by the coordinate method, setting out all five headings.
3. Solve it again by the cosine rule and check the two answers agree.

4. Find a problem in the textbook where the vector method is clearly best, and say why.

Symmetry in three dimensions

Cambridge insert: planes of symmetry and axes of rotational symmetry, on the solids the rest of the year is built from.

LESSON 3 | 1 × 40 MIN | GEOMETRY 10, §2 (EXTENSION) | IGCSE E4.6

LESSON CLOCK

7' | 12' | 11' | 7' | 3'

Plane versus axis	7 min
Counting the planes of a cuboid	12 min
Rotational axes and their order	11 min
Using symmetry to save work	7 min
Homework	3 min

LEARNING OBJECTIVES

- Find the planes of symmetry of a cuboid, prism, pyramid, cylinder and cone.
- Find the axes of rotational symmetry and state each order.
- Distinguish a plane of symmetry from an axis of symmetry.
- Use symmetry to shorten a calculation.

TEXTBOOK

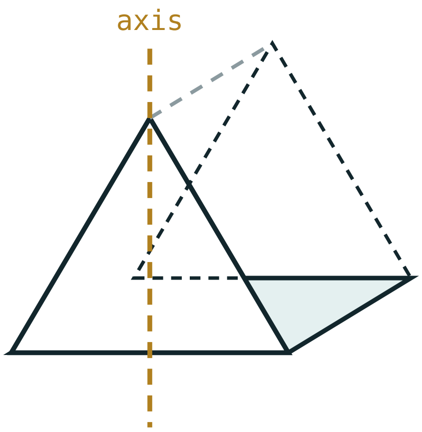
National	pp. 15–18
Cambridge	Core & Extended, pp. 268–273
Workbook	IGCSE Exercise 4.6

01 Explanation

Two different things

PLANE AND AXIS

A **plane of symmetry** slices the solid into two mirror halves. An **axis of rotational symmetry** is a line about which the solid can be turned by less than a full turn and look unchanged; the number of positions is the **order**.



A cuboid cut by one of its three planes of symmetry, and turned about one of its axes.

A solid can have many of each, and the two counts are independent — a cone has infinitely many planes but exactly one axis.

Counting them

SOLID	PLANES	AXES (WITH ORDER)
cuboid, all edges different	*3*	3, each of order 2
square-based cuboid	*5*	1 of order 4, 4 of order 2
cube	*9*	3 of order 4, 4 of order 3, 6 of order 2
triangular prism (equilateral)	*4*	1 of order 3, 3 of order 2
square-based pyramid	*4*	1 of order 4
cylinder	infinitely many, plus 1 horizontal	1 of infinite order
cone	infinitely many	1 of infinite order
sphere	infinitely many	infinitely many

THE CUBOID TRAP

> A cuboid has **3** planes of symmetry, not 6. The two halves either side of a plane must be mirror images — a diagonal cut through a rectangular face does not give that.

Symmetry as a shortcut

Symmetry is not decoration. It halves work:

- The centre of mass lies on every plane of symmetry — so on their intersection.
- A section perpendicular to an axis of order n repeats n times.
- Two lengths symmetric about a plane are equal, with no calculation.

In a cube of side a, the two space diagonals from one face meet on the vertical axis, so their intersection is at height $\frac{a}{2}$ — read straight from symmetry.

02 Worked examples

EXAMPLE 1 **How many planes of symmetry has a cuboid $3 \times 4 \times 5$?**

1. All three edge lengths differ.

   No diagonal plane works.

2. One plane parallel to each pair of opposite faces.

ANSWER

3

EXAMPLE 2 **A prism has a regular hexagonal cross-section. How many planes of symmetry?**

1. The hexagon has 6 lines of symmetry.
   6 vertical planes.
2. One horizontal plane halfway up.

ANSWER

7

EXAMPLE 3 **State the order of rotational symmetry of a square-based pyramid about its vertical axis.**

1. Turning by $90°$ maps the square base to itself.
2. Four positions in a full turn.

ANSWER

Order 4

03 Interactive model

Hold a cuboid box and slice it with a sheet of card to show each plane.

Counting symmetry

A cuboid with all edges different has how many planes of symmetry?

- *2*
- *3*
- *6*
- *9*

A cube has how many planes of symmetry?

- *3*
- *6*
- *9*
- *12*

A cone has how many axes of rotational symmetry?

- *0*
- *1*
- *2*
- infinitely many

A square-based pyramid has rotational order:

- *1*
- *2*
- *4*
- *8*

A regular triangular prism has how many planes?

- *3*
- *4*
- *5*
- *6*

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Plane of symmetry	Simmetriya tekisligi	Плоскость симметрии
Axis of symmetry	Simmetriya o‘qi	Ось симметрии
Rotational symmetry	Aylanma simmetriya	Вращательная симметрия
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Reflection	Akslantirish	Отражение
Mirror image	Ko‘zgu tasviri	Зеркальное изображение
Cuboid	Parallelepiped	Прямоугольный параллелепипед
Cross-section	Kesim	Сечение
Regular solid	Muntazam jism	Правильное тело

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A plane of symmetry produces:

- two mirror halves
- two equal volumes only
- a rotation
- a shadow

The order of rotational symmetry of a cylinder about its axis is:

- *1*
- *2*
- *4*
- infinite

A sphere has how many planes of symmetry?

- *1*
- *3*
- *9*
- infinitely many

A cube has axes of order 3 through:

- face centres
- opposite vertices
- edge midpoints
- it has none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Planes of symmetry of a cube

   ANSWER 9

2. Planes of symmetry of a $2 \times 3 \times 4$ cuboid

   ANSWER 3

3. Planes of symmetry of a sphere

   ANSWER infinitely many

4. Rotational order of a cube about a face axis

   ANSWER 4

5. Axes of a cone

   ANSWER 1

6. Planes of a square-based pyramid

   ANSWER 4

7. Rotational order of a cylinder

   ANSWER infinite

Medium · the standard question

7 PROBLEMS

1. Planes of a regular hexagonal prism

   ANSWER 7

2. Planes of a regular pentagonal pyramid

   ANSWER 5

3. Rotational order of a regular hexagonal prism about its long axis

   ANSWER 6

4. Planes of a cuboid with a square base $3 \times 3 \times 5$

   ANSWER 5

5. Axes of order 2 in a cube

   ANSWER 6

6. Axes of order 3 in a cube

   ANSWER 4

7. Total rotational axes of a cube

   ANSWER 13

Hard · stretch and reason

7 PROBLEMS

1. Explain why a cuboid $a \times a \times b$ with $a \neq b$ has 5 planes

   ANSWER 3 face planes plus 2 diagonal planes of the square cross-section

2. A regular tetrahedron: count its planes

   ANSWER 6

3. A regular tetrahedron: count its rotational axes and orders

   ANSWER 4 of order 3, 3 of order 2

4. Why has a cone infinitely many planes but only one axis?

   ANSWER Every plane through the axis is a mirror; only that line is fixed

5. A solid has 1 plane only. Sketch one

   ANSWER e.g. a scalene-triangle prism cut asymmetrically

6. Use symmetry to find the centre of a cube of side 6 from a corner

   ANSWER $(3, 3, 3)$

7. A cuboid $a \times a \times a$ — why does the count jump from 5 to 9?

   ANSWER The three square cross-sections each contribute two diagonal planes

07 Homework

Homework — 4 tasks

Sketch each solid once and draw the planes on the sketch.

1. Count the planes of symmetry of a cube, a $2 \times 2 \times 5$ cuboid and a regular hexagonal prism.
2. State every rotational axis of a cube with its order, and check the total is 13.
3. Find a solid with exactly two planes of symmetry and sketch it.
4. Explain why a plane of symmetry always divides a solid into equal volumes, but equal volumes do not imply a plane of symmetry.

Basic concepts and axioms of solid geometry

Three axioms about planes, and the two consequences that every later proof in the course will quote.

LESSON 4–5 | 2 × 40 MIN | GEOMETRY 10, §3 | P1 · 3-D CONTEXT

LESSON CLOCK

9' | 16' | 20' | 20' | 11' | 4'

From plane to space	9 min
The three axioms	16 min
Two theorems proved from them	20 min
How many planes?	20 min
Practice	11 min
Homework	4 min

LEARNING OBJECTIVES

- State the three axioms that extend plane geometry into space.
- Prove that a line and a point not on it determine a plane.
- Prove that two intersecting lines determine a plane.
- Decide how many planes a given set of points determines.

TEXTBOOK

National	pp. 19–28
Cambridge	Core & Extended, pp. 274–276
Workbook	Exercise 3.1

01 Explanation

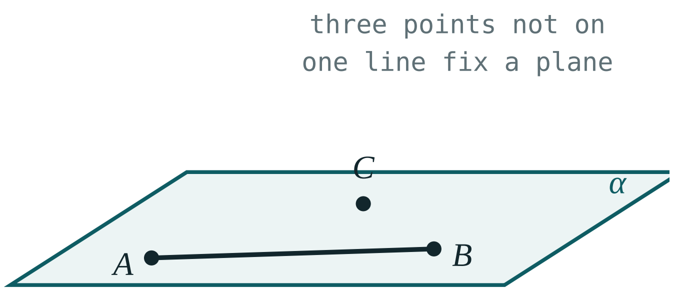
What changes when you leave the plane

In the plane, two lines either meet or are parallel. In space a third case appears: they may be **skew** — neither meeting nor parallel, because they do not lie in a common plane. Every difficulty of solid geometry comes from that one extra case.

> THE NEW UNDEFINED TERM
>
> Plane geometry needed **point** and **line**. Space adds **plane** — flat, unbounded, with no thickness. It is described only by how it behaves.

The three axioms of space

№	AXIOM	WHAT IT FORBIDS
S1	Through any three non-collinear points passes exactly one plane	a tripod wobbling
S2	If two points of a line lie in a plane, the whole line lies in it	a straight ruler lifting off a flat table
S3	If two distinct planes have a common point, they meet in a line through it	two walls meeting at a single point



S1, S2 and S3 in one diagram. Three points fix the plane; the line stays in it; two planes share a whole line.

WHY THREE POINTS, NOT TWO

Two points leave a plane free to spin about the line through them — infinitely many planes contain any given line. The third point, off that line, stops the spin. This is why a three-legged stool never rocks and a four-legged one does.

Two theorems, proved immediately

Theorem 1. A line and a point not on it determine exactly one plane.

Proof. Take two points A, B on the line and the outside point C. They are non-collinear, so by S1 exactly one plane α contains them. By S2 the whole line lies in α. Any plane containing the line and C contains A, B, C, so equals α by S1. ∎

Theorem 2. Two intersecting lines determine exactly one plane.

Proof. Let them meet at P. Choose A on the first and B on the second, both different from P. A, B, P are non-collinear, so S1 gives a unique plane; S2 puts both whole lines inside it. ∎

FOUR WAYS TO FIX A PLANE

three non-collinear points · a line and a point off it · two intersecting lines · two parallel lines. Every one of them reduces to S1.

Counting planes

How many planes are determined by n points, no four coplanar and no three collinear? Every choice of 3 gives one plane:

$$\text{number of planes} = C(n, 3) = \frac{n(n-1)(n-2)}{6}$$

Four points give 4 planes — the four faces of a tetrahedron. Five give 10.

THE CONDITIONS MATTER

If three of the points are collinear they determine no plane by themselves. If all four are coplanar, the four triples give the **same** plane, so the count is 1, not 4.

02 Worked examples

EXAMPLE 1 **How many planes are determined by four points of which exactly three are collinear?**

1. The three collinear points determine a line, not a plane.
2. That line and the fourth point determine one plane.
   Theorem 1.
3. No other triple is non-collinear off it.

ANSWER

1

EXAMPLE 2 **Two planes meet. Can they share exactly two points?**

1. S3: sharing one point forces a whole common line.
2. A line has infinitely many points.

ANSWER

No — they share either nothing, a line, or everything

EXAMPLE 3 **A cube $ABCDA_1B_1C_1D_1$. How many planes are determined by its 8 vertices, taken as faces and diagonal planes?**

1. 6 faces.
2. 6 diagonal planes through opposite edges.
   e.g. ABC_1D_1.
3. Total for these two families.

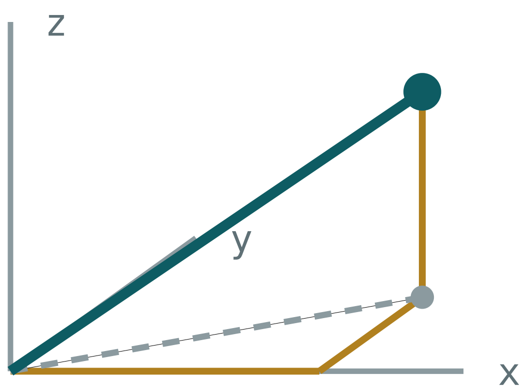
ANSWER

12 (6 faces + 6 diagonal planes)

03 Interactive model

Stand three pencils in plasticine to show a plane fixed, then add a fourth to show it must be adjusted.

A point in space



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Solid geometry (stereometry)	Stereometriya	Стереометрия
Plane	Tekislik	Плоскость
Space	Fazo	Пространство
Collinear points	Bir to‘g‘ri chiziqdagi nuqtalar	Коллинеарные точки
Coplanar points	Bir tekislikdagi nuqtalar	Компланарные точки
Determine a plane	Tekislikni aniqlash	Определять плоскость
Line of intersection	Kesishish chizig‘i	Линия пересечения
Half-space	Yarim fazo	Полупространство
Belongs to a plane	Tekislikka tegishli	Принадлежит плоскости

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Three non-collinear points determine:

- no plane
- one plane
- two planes
- infinitely many

Two points determine how many planes?

- 1
- 2
- 0
- infinitely many

If two planes share a point they share:

- only that point
- a line
- everything
- nothing else

Two lines in space that neither meet nor are parallel are:

- perpendicular
- skew
- coplanar
- impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How many planes through 3 non-collinear points?

   ANSWER *1*

2. How many planes through 1 line?

   ANSWER infinitely many

3. How many planes through 2 intersecting lines?

   ANSWER *1*

4. State axiom S2

   ANSWER Two points of a line in a plane put the whole line in it

5. Can two planes meet in one point only?

   ANSWER no

6. How many planes through 4 points of a tetrahedron?

   ANSWER *4*

7. What is a skew pair of lines?

   ANSWER Neither meeting nor parallel

Medium · the standard question

7 PROBLEMS

1. How many planes from 5 points, no 4 coplanar, no 3 collinear?

   ANSWER *10*

2. How many planes from 6 such points?

   ANSWER *20*

3. Four coplanar points determine how many planes?

   ANSWER *1*

4. Four points, three collinear — how many planes?

   ANSWER *1*

5. Two parallel lines determine how many planes?

   ANSWER *1*

6. Faces of a cube determine how many planes?

   ANSWER *6*

7. Prove two intersecting lines are coplanar

   ANSWER Take a point on each plus the meeting point; apply S1 and S2

Hard · stretch and reason

7 PROBLEMS

1. How many planes are determined by the 8 vertices of a cube in total?

   ANSWER 20

2. Explain why a four-legged table can rock but a three-legged one cannot

   ANSWER Three points always determine a plane; four need not be coplanar

3. Prove: if a line meets a plane in two points it lies in the plane

   ANSWER Directly by S2

4. Three planes pairwise intersecting: how many lines can they give?

   ANSWER 1 or 3

5. Can three planes have exactly one common point?

   ANSWER Yes — like the corner of a room

6. How many planes from n points in general position?

   ANSWER $\frac{n(n-1)(n-2)}{6}$

7. Give a configuration of 5 points determining exactly 5 planes

   ANSWER Four coplanar plus one off the plane

07 Homework

Homework — 5 tasks

Every proof must quote S1, S2 or S3 by name on the line that uses it.

1. State the three axioms of solid geometry from memory.
2. Prove that two parallel lines determine exactly one plane.
3. How many planes do 7 points in general position determine?
4. Explain, with a sketch, why two planes cannot meet in exactly one point.
5. List the four ways of determining a plane and show each reduces to S1.

The mutual positions of lines and planes in space

Every possible arrangement of two lines, of a line and a plane, and of two planes — nine cases in all.

LESSON 6–7 | 2 × 40 MIN | GEOMETRY 10, §4 | IGCSE E4.7

LESSON CLOCK

7' | 14' | 18' | 22' | 12' | 7'

Two lines — three cases	7 min
Line and plane; plane and plane	14 min
Finding skew pairs in a cuboid	18 min
The angle between skew lines	22 min
Practice	12 min
Homework	7 min

LEARNING OBJECTIVES

- List the three possible positions of two lines in space.
- List the three positions of a line and a plane, and the three of two planes.
- Recognise skew lines in a cuboid and name pairs of them.
- Find the angle between two skew lines by translating one of them.

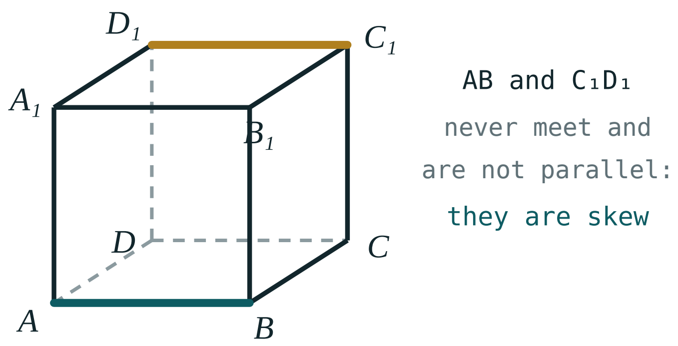
TEXTBOOK

National	pp. 29–40
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 4.1

01 Explanation

Two lines: three cases, not two

CASE	COMMON POINTS	COPLANAR?
intersecting	*1*	yes
parallel	*0*	yes
skew	*0*	no



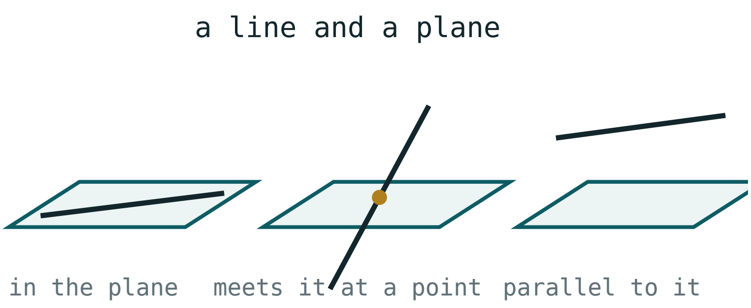
An edge of the top face and a non-touching edge of the bottom face. They never meet, and no plane holds both.

ZERO COMMON POINTS IS NOT ENOUGH

Parallel and skew lines both fail to meet. What separates them is whether a single plane can hold both. Always check for a common plane before writing “parallel”.

A line and a plane, and two planes

OBJECTS	CASE	COMMON POINTS
line and plane	line lies in the plane	infinitely many
	line meets the plane	*1*
	line parallel to the plane	*0*
two planes	coincident	infinitely many
	intersecting	a whole line
	parallel	*0*



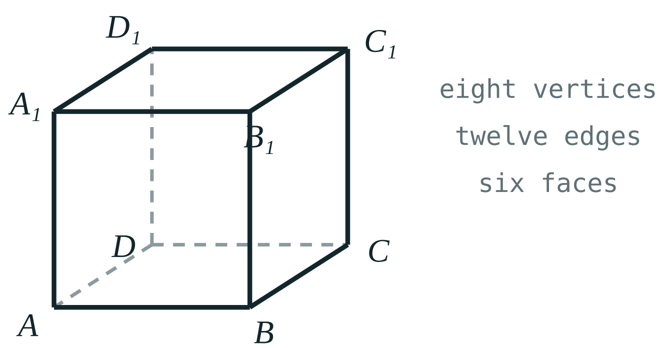
In the plane, meeting the plane at a point, and parallel to it.

> THE TEST FOR A LINE PARALLEL TO A PLANE
>
> A line not in a plane is parallel to it exactly when it is parallel to **some** line lying in that plane. Finding that one line is the whole of most proofs.

Skew pairs in a cuboid

The cuboid $ABCDA_1B_1C_1D_1$ is the standard laboratory. Take edge AB:

RELATION TO AB	EDGES
meets it	AD, BC, AA_1, BB_1
parallel to it	DC, A_1B_1, D_1C_1
skew to it	CC_1, DD_1, A_1D_1, B_1C_1



The standard labelling. Every question about a cube uses these eight letters.

Four skew, four meeting, three parallel — eleven other edges, and every one accounted for.

The angle between skew lines

THE METHOD

Skew lines never meet, so there is no angle to measure directly. **Translate one of them** until it passes through a point of the other; the angle between the resulting intersecting lines is defined to be the angle between the skew pair. It does not depend on the point chosen.

In a cube of side a, find the angle between AB and B_1C_1. Translate B_1C_1 down to BC. Now $AB \perp BC$, so the answer is $90°$ — skew lines can be perpendicular without ever meeting.

$$\text{angle between } AB \text{ and } DC_1\text{: translate } DC_1 \text{ to } AB_1\text{, then } \tan\theta = \frac{a}{a} = 1 \Rightarrow \theta = 45°$$

02 Worked examples

EXAMPLE 1 **In cube $ABCDA_1B_1C_1D_1$, name all edges skew to AA_1.**

1. Edges meeting AA_1: AB, AD, A_1B_1, A_1D_1.
2. Edges parallel: BB_1, CC_1, DD_1.
3. The rest are skew.

ANSWER

BC, CD, B_1C_1, C_1D_1

EXAMPLE 2 **Cube of side a. Find the angle between AB_1 and BC_1.**

1. Both are face diagonals, length $a\sqrt{2}$.
2. Translate BC_1 to AD_1 — now they meet at A.
3. $B_1D_1 = a\sqrt{2}$

   Triangle AB_1D_1 is equilateral.
4. All sides $a\sqrt{2}$.

ANSWER

$60°$

EXAMPLE 3 **Two planes are each parallel to a line ℓ. Must they be parallel to each other?**

1. Consider two walls of a room and the vertical line where the floor meets nothing.
2. Both can be parallel to a vertical line and still meet.

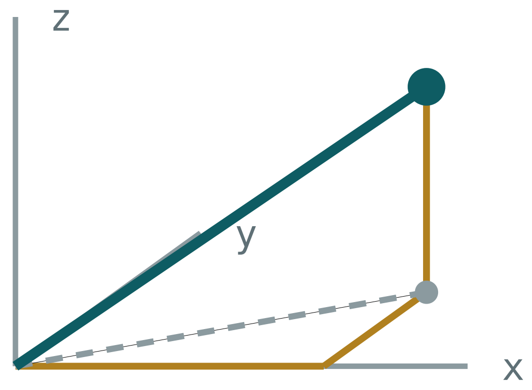
ANSWER

No — they may intersect in a line parallel to ℓ

03 Interactive model

Hold two pencils as skew lines, then slide one until they meet, keeping its direction.

Lines in a box



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Skew lines	Ayqash to‘g‘ri chiziqlar	Скрещивающиеся прямые
Intersecting lines	Kesishuvchi to‘g‘ri chiziqlar	Пересекающиеся прямые
Parallel lines	Parallel to‘g‘ri chiziqlar	Параллельные прямые
Line parallel to a plane	Tekislikka parallel chiziq	Прямая, параллельная плоскости
Line in a plane	Tekislikdagi chiziq	Прямая, лежащая в плоскости
Intersecting planes	Kesishuvchi tekisliklar	Пересекающиеся плоскости
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Angle between skew lines	Ayqash chiziqlar orasidagi burchak	Угол между скрещивающимися прямыми
Common perpendicular	Umumiy perpendikulyar	Общий перпендикуляр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Skew lines are:

- parallel
- intersecting
- non-coplanar and non-intersecting
- perpendicular

A line and a plane can share:

- 0, 1 or infinitely many points
- only 0 or 1
- only 1
- exactly 2

Two distinct planes can share:

- a point
- a line
- two points only
- nothing but a point

The angle between skew lines is found by:

- measuring the gap
- translating one to meet the other
- projecting onto the floor
- it is undefined

In a cube, how many edges are skew to a given edge?

- 2
- 3
- 4
- 6

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Name the three positions of two lines in space

ANSWER intersecting, parallel, skew

2. Name the three positions of a line and a plane

ANSWER in it, meeting it, parallel to it

3. Name the three positions of two planes

ANSWER coincident, intersecting, parallel

4. In a cube, is AB parallel to D_1C_1?

ANSWER yes

5. In a cube, is AB skew to CC_1?

ANSWER yes

6. How many edges of a cube meet AB?

ANSWER 4

7. Can skew lines be perpendicular?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. List all edges skew to BC in a cube

ANSWER $AA_1, DD_1, A_1B_1, C_1D_1$

2. Angle between AB and B_1C_1 in a cube

ANSWER $90°$

3. Angle between AB and DC_1 in a cube

ANSWER $45°$

4. Angle between AB_1 and BC_1 in a cube

ANSWER $60°$

5. Two planes parallel to the same line — must they be parallel?

ANSWER no

6. Two lines parallel to the same plane — must they be parallel?

ANSWER no

7. Two planes parallel to the same plane — must they be parallel?

ANSWER yes

Hard · stretch and reason

7 PROBLEMS

1. Cuboid $4 \times 3 \times 12$. Angle between AB and DB_1

   ANSWER $\approx 71.9°$

2. Prove that if $a \parallel b$ and $b \parallel c$ in space then $a \parallel c$

   ANSWER Transitivity of direction

3. Give two skew lines that are perpendicular in a cube

   ANSWER AB and B_1C_1

4. How many pairs of skew edges has a cube?

   ANSWER 24

5. Prove a line parallel to two intersecting planes is parallel to their line of intersection

   ANSWER It is parallel to a line in each; the common direction is the intersection

6. Find the common perpendicular of AB and CC_1 in a unit cube

   ANSWER BC, length 1

7. Explain why the angle between skew lines does not depend on where you translate

   ANSWER Translation preserves direction

07 Homework

Homework — 5 tasks

Draw the cube once and answer tasks 1–3 from that one drawing.

1. For cube $ABCDA_1B_1C_1D_1$, list the edges meeting, parallel to and skew to CD.
2. Find the angle between AD and BC_1.
3. Find the angle between the face diagonals AC and BD_1.
4. Give an example showing two planes parallel to the same line need not be parallel.
5. Explain in three sentences the difference between parallel and skew.

Spatial figures. Polyhedra

The vocabulary of solids — face, edge, vertex — and the one formula that ties all three together.

LESSON 8–9 | 2 × 40 MIN | GEOMETRY 10, §5 | IGCSE E4.8

LESSON CLOCK

8' | 14' | 18' | 14' | 16' | 10'

The three words	8 min
Prisms and pyramids	14 min
Euler's formula	18 min
The five regular solids	14 min
Practice	16 min
Homework	10 min

LEARNING OBJECTIVES

- Name the faces, edges and vertices of a polyhedron and count them.
- Verify Euler's formula for a range of solids.
- Distinguish convex from non-convex polyhedra.
- Name the five regular polyhedra and their face shapes.

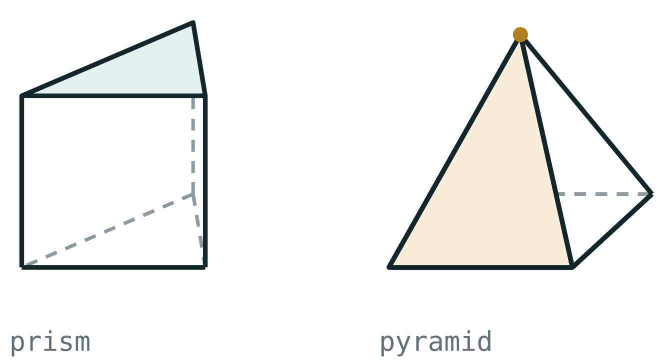
TEXTBOOK

National	pp. 41–52
Cambridge	Core & Extended, pp. 282–288
Workbook	Exercise 5.1

01 Explanation

Face, edge, vertex

A **polyhedron** is a solid bounded by flat polygons. Those polygons are its **faces**; where two faces meet is an **edge**; where edges meet is a **vertex**.



A prism and a pyramid, with one face, one edge and one vertex marked on each.

SOLID	F	V	E
cube	6	8	12
triangular prism	5	6	9
square pyramid	5	5	8
tetrahedron	4	4	6
hexagonal prism	8	12	18
octahedron	8	6	12

Prisms and pyramids

TWO FAMILIES

A **prism** has two identical parallel bases joined by parallelograms. A **pyramid** has one base and all lateral edges meeting at a single apex.

For an n-gonal solid the counts follow from the base:

SOLID	F	V	E
n-gonal prism	$n + 2$	$2n$	$3n$
n-gonal pyramid	$n + 1$	$n + 1$	$2n$

A prism is **right** when the lateral edges are perpendicular to the bases, otherwise **oblique**. A pyramid is **regular** when its base is a regular polygon and the apex sits above the base centre.

Euler’s formula

FOR EVERY CONVEX POLYHEDRON

$$F + V - E = 2$$

Faces plus vertices, minus edges, is always two — whatever the solid, however many faces.

Check the table above: $6 + 8 - 12 = 2$, $5 + 6 - 9 = 2$, $4 + 4 - 6 = 2$. It never fails for a convex solid.

The formula is a **tool**, not a curiosity: it finds a missing count. A solid with 12 pentagonal faces has $E = \dfrac{12 \times 5}{2} = 30$ edges, so $V = 2 - F + E = 2 - 12 + 30 = 20$.

CONVEX ONLY

A polyhedron with a hole through it — a picture frame, say — gives $F + V - E = 0$. Euler's formula assumes the surface can be deformed onto a sphere.

The five regular polyhedra

A polyhedron is **regular** when all faces are congruent regular polygons and the same number of them meet at every vertex. There are exactly five — a fact known to the Greeks:

NAME	FACES	F	V	E
tetrahedron	4 triangles	4	4	6
cube (hexahedron)	6 squares	6	8	12
octahedron	8 triangles	8	6	12
dodecahedron	12 pentagons	12	20	30
icosahedron	20 triangles	20	12	30

Why only five: at a vertex at least three faces must meet and their angles must total less than $360°$. Triangles ($60°$) allow 3, 4 or 5; squares ($90°$) allow only 3; pentagons ($108°$) only 3; hexagons ($120°$) give exactly $360°$ and lie flat. Three plus one plus one — five solids, and no more are possible.

02 Worked examples

EXAMPLE 1 **A polyhedron has 9 faces and 14 vertices. How many edges?**

1. $F + V - E = 2$
2. $9 + 14 - E = 2$
3. $E = 21$

ANSWER

21

EXAMPLE 2 **A pentagonal prism: find *F, V, E* and check Euler.**

1. $n = 5$: $F = 7$, $V = 10$, $E = 15$.
2. $7 + 10 - 15 = 2$ ✓

ANSWER

$F = 7, V = 10, E = 15$

EXAMPLE 3 **A solid has only triangular faces and 12 vertices, with 5 faces at each vertex. Find F and E.**

1. Each vertex has 5 edges; each edge has 2 ends.

$$E = \frac{12 \times 5}{2} = 30$$

2. $F = 2 - V + E = 2 - 12 + 30 = 20$

3. This is the icosahedron.

ANSWER

$F = 20, E = 30$

03 Interactive model

Count F, V and E on a real box, then on a pyramid net, and check Euler each time.

Counting a polyhedron

A cube has how many edges?

- 6
- 8
- 12
- 16

A hexagonal pyramid has how many faces?

- 6
- 7
- 8
- 12

$F = 8, E = 12$ gives V equal to:

- 4
- 6
- 8
- 10

How many regular polyhedra are there?

- 3
- 4
- 5
- infinitely many

A dodecahedron has faces that are:

- triangles
- squares
- pentagons
- hexagons

Why is there no regular solid with hexagonal faces?

- angles sum to $360°$
- hexagons are not regular
- too many edges
- there is one

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Polyhedron	Ko‘pyoq	Многогранник
Face	Yoq	Грань
Edge	Qirra	Ребро
Vertex	Uchi	Вершина
Convex polyhedron	Qavariq ko‘pyoq	Выпуклый многогранник
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Regular polyhedron	Muntazam ko‘pyoq	Правильный многогранник
Euler’s formula	Eyler formulasi	Формула Эйлера
Lateral face	Yon yoq	Боковая грань
Base	Asos	Основание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Euler's formula is:

- $F + V + E = 2$
- $F + V - E = 2$
- $F - V + E = 2$
- $F \cdot V = E$

A triangular prism has:

- $F=5, V=6, E=9$
- $F=6, V=5, E=9$
- $F=5, V=9, E=6$
- $F=4, V=4, E=6$

A pyramid on an n-gon has how many edges?

- n
- $n + 1$
- $2n$
- $3n$

Where two faces meet is:

- a vertex
- an edge
- a face
- a base

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. F, V, E of a cube

ANSWER $6, 8, 12$

2. F, V, E of a tetrahedron

ANSWER $4, 4, 6$

3. F, V, E of a square pyramid

ANSWER $5, 5, 8$

4. Check Euler for a cube

ANSWER $6 + 8 - 12 = 2$

5. Faces of an octahedron

ANSWER 8 triangles

6. Faces of a hexagonal prism

ANSWER 8

7. How many regular polyhedra?

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. $F = 10$, $V = 16$; find E

ANSWER 24

2. $V = 12$, $E = 18$; find F

ANSWER 8

3. F, V, E of a heptagonal prism

ANSWER $9, 14, 21$

4. F, V, E of a decagonal pyramid

ANSWER $11, 11, 20$

5. A solid has 20 triangular faces. How many edges?

ANSWER 30

6. And how many vertices?

ANSWER 12

7. Which regular solid is that?

ANSWER icosahedron

Hard · stretch and reason

7 PROBLEMS

1. A solid has 12 pentagonal and 20 hexagonal faces, 3 at each vertex. Find V and E

   ANSWER $V = 60, E = 90$

2. Show a polyhedron cannot have exactly 7 edges

   ANSWER $F + V = 9$ forces an impossible face count

3. Prove $2E \geq 3F$ for any polyhedron

   ANSWER Every face has at least 3 edges, each shared by 2 faces

4. Prove $2E \geq 3V$

   ANSWER Every vertex meets at least 3 edges

5. Use both to show $F \leq 2V - 4$

   ANSWER Substitute into Euler

6. A prism and a pyramid have the same number of edges. Relate their bases

   ANSWER $3n = 2k$ — e.g. a hexagonal pyramid and a quadrilateral prism

7. Explain why a picture-frame solid fails Euler

   ANSWER Its surface is a torus, not a sphere

07 Homework

Homework — 5 tasks

Task 5 needs the angle argument, not just the number five.

1. Complete a table of F, V, E for a cube, a triangular prism, a pentagonal pyramid and an octahedron, and check Euler on each row.
2. A polyhedron has 15 faces and 25 edges. Find its number of vertices.
3. Write the formulas for F, V, E of an n-gonal prism and pyramid.
4. Name the five regular polyhedra with their face shapes and counts.

5. Explain why no regular polyhedron can have hexagonal faces.

Home / Grade 10 / Geometry / Quarter 1

Making models of polyhedra

A practical pair of lessons: from net to solid, and what a badly folded model teaches that a correct one does not.

LESSON 10–11 | 2 × 40 MIN | GEOMETRY 10, §6 | IGCSE E4.8

LESSON CLOCK

7' | 14' | 28' | 14' | 12' | 5'

What a net must satisfy	7 min
The eleven nets of a cube	14 min
Building — prism, pyramid, tetrahedron	28 min
Surface area from the net	14 min
Checking the models	12 min
Homework	5 min

LEARNING OBJECTIVES

- Draw an accurate net of a prism, a pyramid and a tetrahedron.
- Build the solid from the net and check its face, edge and vertex counts.
- Explain why some arrangements of squares do not fold into a cube.
- Compute the surface area from the net.

TEXTBOOK

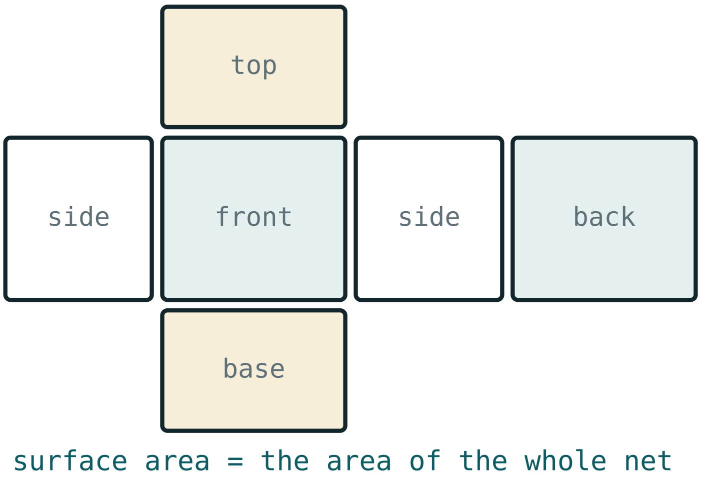
National	pp. 53–60
Cambridge	Core & Extended, pp. 289–293
Workbook	Net templates N1–N5

01 Explanation

What a net has to satisfy

A **net** is the surface of a solid cut along some edges and laid flat. To fold back it must satisfy three conditions:

1. Exactly the right faces, each the right size.
2. Edges that will be joined must be equal in length.
3. The arrangement must not force two faces to overlap.



A net and the solid it folds to. Matching edges are the same length.

CONDITION 3 IS THE ONE PEOPLE FORGET

Six squares in a row have the right faces and the right edge lengths and still do not make a cube — the ends meet before the sides close.

The eleven nets of a cube

Of the 35 ways to join six squares edge to edge, exactly **eleven** fold into a cube.
They divide by the length of their longest row:

LONGEST ROW	HOW MANY	DESCRIPTION
4	*6*	a row of four, one square above and one below
3	*4*	two rows, of three and three, offset
2	*1*	a staircase of three pairs

A QUICK TEST

In a cube net, no 2×2 block of squares can appear — such a block would fold two faces onto each other. Any arrangement containing one is not a cube net.

Building, and computing area from the net

Three models, in pairs. Card, ruler, compasses, 5 mm tabs on alternate edges.

MODEL	NET	SURFACE AREA
cube, side a	6 squares	$6a^2$
n-gonal prism	2 bases + a rectangle $P \times h$	$2B + Ph$
regular pyramid	base + n triangles	$B + \frac{1}{2}Pl$
regular tetrahedron, edge a	4 equilateral triangles	$\sqrt{3}\,a^2$

Here B is the base area, P the base perimeter, h the height and l the slant height. The net makes each formula obvious: the lateral surface of a prism is literally one rectangle.

SLANT HEIGHT IS NOT HEIGHT

For a pyramid the triangles use l, the distance from the apex to the midpoint of a base edge — not the vertical height h. They are related by $l^2 = h^2 + (\frac{\text{base edge}}{2})^2$ for a square base.

02 Worked examples

EXAMPLE 1 **A square-based pyramid has base 10 cm and height 12 cm. Find its surface area.**

1. $l^2 = 12^2 + 5^2 = 169$

   Slant height.

2. $l = 13$

3. Base 100; four triangles $4 \times \frac{1}{2} \times 10 \times 13 = 260$.

4. $100 + 260 = 360$

ANSWER

$360\ cm^2$

EXAMPLE 2 **A triangular prism has an equilateral base of side 6 cm and length 15 cm. Find its surface area.**

1. Base area $\frac{\sqrt{3}}{4} \times 36 = 9\sqrt{3} \approx 15.59$
2. Two bases ≈ 31.18.
3. Lateral: perimeter $18 \times 15 = 270$.
4. Total ≈ 301.2.

ANSWER

$\approx 301\ cm^2$

EXAMPLE 3 **Explain why six squares in a single row cannot fold into a cube.**

1. Folding the row wraps four squares round the sides.
2. The fifth returns to the first face's position.
   Overlap.
3. Two faces are left uncovered.

ANSWER

It violates condition 3 — faces overlap and two are missing

03 Interactive model

Have each pair test one candidate cube net on squared paper before cutting card.

Will it fold?

How many nets of a cube are there?

- 6
- 8
- 11
- 35

A net containing a 2×2 block of squares:

- folds fine
- never folds to a cube
- makes a cuboid
- makes a pyramid

Surface area of a cube of side 5 cm:

- $25\ cm^2$
- $125\ cm^2$
- $150\ cm^2$
- $100\ cm^2$

The lateral surface of a prism unfolds to:

- a triangle
- one rectangle
- a circle
- a trapezium

A regular tetrahedron’s net is:

- 3 triangles
- 4 triangles
- 4 squares
- 6 triangles

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Net	Yoyma	Развёртка
Fold line	Buklash chizig‘i	Линия сгиба
Tab (flap)	Yopishtirish qismi	Клапан
Surface area	To‘la sirt yuzasi	Площадь полной поверхности
Lateral surface area	Yon sirt yuzasi	Площадь боковой поверхности
Slant height	Apofema	Апофема
Scale of a model	Model masshtabi	Масштаб модели
Template	Andoza	Шаблон
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A net is:

- the volume
- the flattened surface
- a cross-section
- a projection

The slant height of a pyramid is measured to:

- the base centre
- the midpoint of a base edge
- a base vertex
- the apex

Surface area of an n-gonal prism:

- Ph
- $2B + Ph$
- $B + \frac{1}{2}Pl$
- Bh

Six squares in a row fold into:

- a cube
- nothing — they overlap
- a cuboid
- an open box

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Surface area of a cube of side 4 cm

   ANSWER *96 cm²*

2. Surface area of a cube of side 10 cm

   ANSWER *600 cm²*

3. How many faces in a cube net?

   ANSWER *6*

4. How many nets of a cube?

   ANSWER *11*

5. Net of a tetrahedron consists of

   ANSWER *4* equilateral triangles

6. A prism's lateral net is

   ANSWER one rectangle

7. Slant height of a pyramid with $h = 4$, base edge 6

   ANSWER *5*

Medium · the standard question

7 PROBLEMS

1. Square pyramid, base 8, height 3. Surface area

   ANSWER *144 cm²*

2. Cuboid $3 \times 4 \times 5$. Surface area

   ANSWER *94 cm²*

3. Triangular prism, right base 3-4-5, length 10. Surface area

   ANSWER *132 cm²*

4. Regular tetrahedron edge 6. Surface area

   ANSWER $36\sqrt{3} \approx 62.35$ cm²

5. Hexagonal prism, side 2, height 9. Lateral area

   ANSWER *108*

6. Cube surface area is 216 cm². Find the edge

   ANSWER *6 cm*

7. Square pyramid, base 12, slant height 10. Surface area

   ANSWER *384 cm²*

Hard · stretch and reason

7 PROBLEMS

1. A cube net has a 2×2 block. Prove it cannot fold

   ANSWER Two of the four would land on the same face

2. Square pyramid, base 14, surface area 896. Find the height

   ANSWER 24

3. A cuboid has surface area 208 and edges in the ratio $1:2:3$

   ANSWER $4 \times 8 \times 12$? check: $S = 352$ — so ratio gives $x \approx 3.1$

4. Regular tetrahedron of surface area $16\sqrt{3}$. Find the edge

   ANSWER 4

5. How many nets has a regular tetrahedron?

   ANSWER 2

6. A prism has surface area $2B + Ph$. What happens as $h \to 0$?

   ANSWER It flattens to twice the base

7. Design a net for an oblique prism and say why it is harder

   ANSWER The lateral faces are parallelograms of differing shapes

07 Homework

Homework — 4 tasks

Bring the three finished models to the next lesson — they are used for sections.

1. Draw and cut two different nets of a cube, and fold both.
2. Draw the net of a square pyramid with base 8 cm and height 3 cm, and find its surface area.
3. Draw the net of a regular tetrahedron of edge 6 cm and find its surface area.
4. Find an arrangement of six squares that is not a cube net and explain in one sentence why it fails.

Simple sections of polyhedra

Cutting a solid with a plane — the three rules that make the cut, and what shape appears.

LESSON 12–13 | 2 × 40 MIN | GEOMETRY 10, §7 | IGCSE E4.8

LESSON CLOCK

8' | 16' | 19' | 14' | 13' | 10'

What a section is	8 min
The three construction rules	16 min
Sections of a cube	19 min
Areas of sections	14 min
Practice	13 min
Homework	10 min

LEARNING OBJECTIVES

- Construct the section of a cube or prism through three given points.
- Apply the rule that a plane cuts two parallel faces in parallel lines.
- Name the polygon obtained and justify the number of sides.
- Compute the area of a simple section.

TEXTBOOK

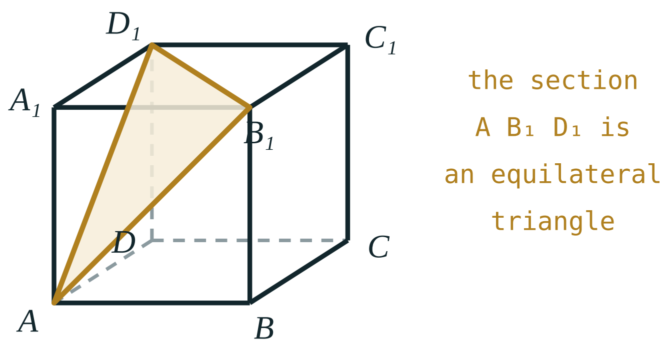
National	pp. 61–70
Cambridge	Core & Extended, pp. 294–298
Workbook	Exercise 7.1

01 Explanation

The three rules

EVERYTHING FOLLOWS FROM THESE

R1 Two points of the cutting plane on the same face join by a straight segment in that face. **R2** A plane cuts two parallel faces in two **parallel** lines. **R3** The section is a closed polygon; every side lies in a face of the solid.



A plane through three points of a cube. R2 forces the top and bottom cuts to be parallel.

R2 does most of the work. Once one side of the section is drawn on a face, the side on the opposite face is fixed in direction — only its position remains to be found.

Sections of a cube

A plane can cut a cube in a triangle, a quadrilateral, a pentagon or a hexagon — but never in a heptagon, because a cube has only six faces and each contributes at most one side.

SECTION	THROUGH	AREA FOR SIDE A
square	parallel to a face	a^2
rectangle	a diagonal plane ABC_1D_1	$a^2\sqrt{2}$
equilateral triangle	three vertices next to one corner	$\frac{\sqrt{3}}{2}a^2$
regular hexagon	midpoints of six edges	$\frac{3\sqrt{3}}{4}a^2$

THE HEXAGON IS THE BEAUTIFUL ONE

Cut a cube through the midpoints of six edges, perpendicular to a space diagonal, and the section is a **regular** hexagon of side $\frac{a\sqrt{2}}{2}$. It is worth building once in card.

Constructing a section, step by step

Problem. Cube $ABCDA_1B_1C_1D_1$. Construct the section through A, C and B_1.

1. A and C are both on the bottom face: draw AC (R1).
2. A and B_1 are both on face ABB_1A_1: draw AB_1 (R1).
3. C and B_1 are both on face BCC_1B_1: draw CB_1 (R1).
4. The polygon closes.

All three segments are face diagonals of length $a\sqrt{2}$, so the section is an **equilateral triangle** of area $\dfrac{\sqrt{3}}{4} \cdot 2a^2 = \dfrac{\sqrt{3}}{2} a^2$.

WHEN TWO POINTS ARE NOT ON ONE FACE

Extend an existing edge of the section until it meets the plane of a face, producing a new point you can use. That extended line is the **trace** of the cutting plane on that face.

02 Worked examples

EXAMPLE 1 **Cube of side 6. Find the area of the diagonal section ABC_1D_1.**

1. $AB = 6$

   An edge.

2. $BC_1 = 6\sqrt{2}$

   A face diagonal.

3. The section is a rectangle.

4. $6 \times 6\sqrt{2} = 36\sqrt{2} \approx 50.9$

ANSWER

$36\sqrt{2} \approx 50.9$ square units

EXAMPLE 2 **Cube of side *a*. Find the area of the hexagonal section through six edge midpoints.**

1. Each side joins midpoints of adjacent edges.

   Side $\frac{a\sqrt{2}}{2}$.

2. The hexagon is regular.

   $S = \frac{3\sqrt{3}}{2} s^2$

3. $s^2 = \frac{a^2}{2}$

4. $S = \frac{3\sqrt{3}}{2} \cdot \frac{a^2}{2} = \frac{3\sqrt{3}}{4} a^2$

ANSWER

$\frac{3\sqrt{3}}{4} a^2 \approx 1.30a^2$

EXAMPLE 3 **What is the maximum number of sides a section of a cube can have, and why?**

1. Each side lies in one face.

   A cube has six faces.

2. A plane meets each face in at most one segment.

3. Six is attained by the midpoint hexagon.

ANSWER

6

03 Interactive model

Slice a plasticine cube with a wire to show the triangle, the rectangle and the hexagon.

Naming a section

A plane parallel to a face of a cube cuts a:

- triangle
- square
- hexagon
- pentagon

The section ABC_1D_1 of a cube is a:

- square
- rectangle
- rhombus
- trapezium

The maximum number of sides of a section of a cube:

- 4
- 5
- 6
- 8

Two parallel faces are cut by a plane in:

- perpendicular lines
- parallel lines
- a point
- nothing

The section through three vertices adjacent to one corner is:

- a right triangle
- an equilateral triangle
- a square
- a hexagon

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Section (cross-section)	Kesim	Сечение
Cutting plane	Kesuvchi tekislik	Секущая плоскость
Trace of a plane	Tekislik izi	След плоскости
Diagonal section	Diagonal kesim	Диагональное сечение
Parallel faces	Parallel yoqlar	Параллельные грани
Hexagonal section	Olti burchakli kesim	Шестиугольное сечение
Construction of a section	Kesim yasash	Построение сечения
Midpoint of an edge	Qirra o‘rtasi	Середина ребра
Regular hexagon	Muntazam olti burchak	Правильный шестиугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A section is:

- a shadow
- the figure where a plane cuts a solid
- a net
- a projection

Rule R2 says a plane cuts parallel faces in:

- equal lines
- parallel lines
- perpendicular lines
- points

The area of the diagonal section of a cube of side a is:

- a^2
- $a^2\sqrt{2}$
- $2a^2$
- $a^2\sqrt{3}$

A cube section can never be a:

- triangle
- pentagon
- hexagon
- heptagon

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Section of a cube parallel to a face

   ANSWER a square

2. Area of that section, side 5

   ANSWER 25

3. Diagonal section of a cube of side 4 — its area

   ANSWER $16\sqrt{2}$

4. Maximum sides of a cube section

   ANSWER 6

5. Section through A, C, B_1 is

   ANSWER an equilateral triangle

6. What does R2 say?

   ANSWER Parallel faces give parallel cuts

7. Can a cube section be a regular pentagon?

   ANSWER no

Medium · the standard question

7 PROBLEMS

1. Cube side 6; area of the section through A, C, B_1

ANSWER $18\sqrt{3} \approx 31.2$

2. Cube side a; area of the midpoint hexagon

ANSWER $\frac{3\sqrt{3}}{4}a^2$

3. Cube side 4; area of the midpoint hexagon

ANSWER $12\sqrt{3} \approx 20.8$

4. Section of a triangular prism parallel to the base

ANSWER a congruent triangle

5. Section of a square pyramid parallel to the base

ANSWER a smaller square

6. Cube side 10; diagonal section area

ANSWER $100\sqrt{2} \approx 141.4$

7. Section of a cube through 5 faces is a

ANSWER pentagon

Hard · stretch and reason

7 PROBLEMS

1. Cube $ABCDA_1B_1C_1D_1$ side 6. Construct and find the area of the section through A, midpoint of BB_1, and C

   ANSWER A rhombus of area $18\sqrt{5} \approx 40.2$

2. Prove the midpoint section of a cube is a regular hexagon

   ANSWER All six sides equal $\frac{a\sqrt{2}}{2}$ by symmetry about the space diagonal

3. Why is that hexagon perpendicular to a space diagonal?

   ANSWER The diagonal is an axis of order 3 through the hexagon centre

4. Cube side a. Find the largest possible area of a plane section

   ANSWER $a^2\sqrt{2}$ — the diagonal rectangle

5. Section of a regular tetrahedron through the midpoints of four edges

   ANSWER a square

6. Prove that section is a square

   ANSWER Each side is a midline, half an opposite edge, and the pairs are perpendicular

7. A plane cuts a cube in a pentagon. How many faces does it miss?

   ANSWER 1

07 Homework

Homework — 5 tasks

Constructions must show the rule used at each step (R1, R2 or R3).

1. Construct the section of a cube through A, C and D_1 and name the shape.
2. Cube of side 8: find the area of the diagonal section.
3. Construct the regular hexagonal section and find its area for side 8.
4. Explain why a section of a cube cannot have seven sides.

5. Construct the section of a square pyramid through the midpoints of two base edges and the apex.

The sine and cosine rules in three dimensions

Cambridge insert: choosing a triangle inside a solid, drawing it flat, and solving it.

LESSON 14–16 | 3 × 40 MIN | GEOMETRY 10, §7 (EXTENSION) | IGCSE E6.5 · P1 · 5.5

LESSON CLOCK

9' | 18' | 23' | 23' | 35' | 12'

Recall the two rules	9 min
Choosing the right triangle	18 min
Angle between a line and a plane	23 min
Angle between two planes	23 min
Practice	35 min
Homework	12 min

LEARNING OBJECTIVES

- Identify a triangle inside a solid that contains the required length or angle.
- Redraw that triangle flat, with all known data marked.
- Apply the sine rule, the cosine rule and the area formula in a 3-D context.
- Find the angle between a line and a plane and between two planes.

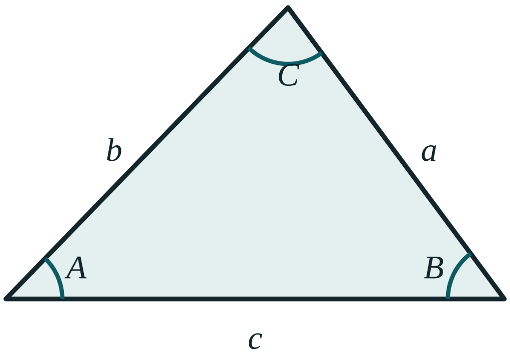
TEXTBOOK

National	pp. 71–78
Cambridge	Core & Extended, pp. 300–312
Workbook	IGCSE Exercise 6.5

01 Explanation

The two rules, recalled

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad a^2 = b^2 + c^2 - 2bc\cdot\cos A$$



$$a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

*The labelling convention: side a opposite angle A.
Every rule assumes it.*

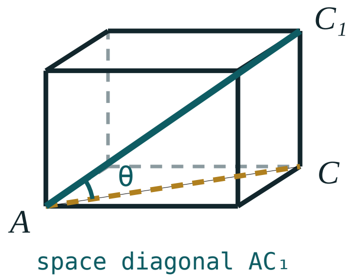
GIVEN	USE
two angles and a side	sine rule
two sides and a non-included angle	sine rule (watch the ambiguous case)
two sides and the included angle	cosine rule
three sides	cosine rule, rearranged for $\cos A$

Area: $S = \frac{1}{2} ab \cdot \sin C$ — two sides and the angle between them.

The method in three dimensions

FOUR STEPS, ALWAYS THE SAME

1 Find a triangle containing what you want. **2** Redraw it flat, large, on its own. **3** Mark every length and angle you already know. **4** Solve it as an ordinary plane triangle.



The box, and beside it the single triangle pulled out and drawn flat.

Almost every 3-D trigonometry question is a plane question hiding inside a solid. The hard part is step 1, and it is made easier by drawing the solid large and marking the right angles first — a vertical edge is perpendicular to **every** line in the base.

The angle between a line and a plane

DEFINITION

Drop a perpendicular from a point of the line to the plane. Join the foot to the point where the line meets the plane. The angle between the line and that **projection** is the angle required — and it is the smallest angle the line makes with any line of the plane.

In a cuboid $a \times b \times c$, the angle θ between the space diagonal and the base:

$$\tan\theta = \frac{c}{\sqrt{a^2 + b^2}}$$

For a $3 \times 4 \times 12$ box: the base diagonal is 5, so $\tan\theta = \dfrac{12}{5}$ and $\theta \approx 67.4°$.

The angle between two planes

Find the line where the planes meet. From a point on it, draw a line in each plane **perpendicular to that line of intersection**. The angle between those two lines is the dihedral angle.

BOTH LINES MUST BE PERPENDICULAR TO THE EDGE

> Any other pair gives the wrong angle. Choosing the edge and then the two perpendiculars is the whole of the technique.

Square pyramid, base 10, height 12: the angle between a lateral face and the base uses the slant height 13 against the apothem 5, giving $\cos\theta = \frac{5}{13}$, so $\theta \approx 67.4°$.

02 Worked examples

EXAMPLE 1 **Cuboid $6 \times 8 \times 10$. Find the length of the space diagonal and its angle with the base.**

1. Base diagonal $\sqrt{36 + 64} = 10$
2. Space diagonal $\sqrt{100 + 100} = 10\sqrt{2} \approx 14.14$
3. $\tan\theta = \frac{10}{10} = 1$
4. $\theta = 45°$

ANSWER

$10\sqrt{2} \approx 14.1; 45°$

EXAMPLE 2 **A pyramid has a square base of side 8 and lateral edge 10. Find the angle between a lateral edge and the base.**

1. Half the base diagonal: $\frac{8\sqrt{2}}{2} = 4\sqrt{2} \approx 5.657$

2. Right triangle: hypotenuse 10, adjacent 5.657.

3. $\cos\theta = \frac{5.657}{10} = 0.5657$

4. $\theta \approx 55.5°$

ANSWER

$\approx 55.5°$

EXAMPLE 3 **In triangle ABC inside a solid, $AB = 7, AC = 9, \angle BAC = 52°$. Find BC and the area.**

1. $BC^2 = 49 + 81 - 2·7·9·\cos 52°$
   Cosine rule.

2. $= 130 - 126 \times 0.6157 = 52.42$

3. $BC \approx 7.24$

4. $S = \frac{1}{2} \cdot 7 \cdot 9 \cdot \sin 52° \approx 24.8$

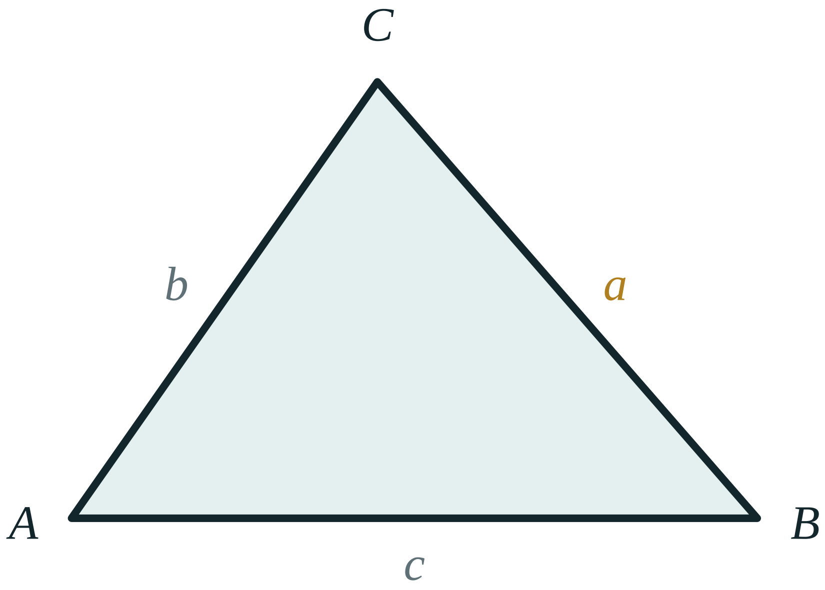
ANSWER

$BC \approx 7.24$, area ≈ 24.8

03 Interactive model

Hold a box and point at the triangle before any calculation begins.

Sine and cosine rules



b **7** | c **9** | A **55°** | a **7.6** | B **49°** | C **76°** | area **25.8**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sine rule	Sinuslar teoremasi	Теорема синусов
Cosine rule	Kosinuslar teoremasi	Теорема косинусов
Included angle	Ichki burchak	Угол между сторонами
Angle between a line and a plane	To‘g‘ri chiziq va tekislik orasidagi burchak	Угол между прямой и плоскостью
Angle between two planes	Ikki tekislik orasidagi burchak	Угол между плоскостями
Projection	Proyeksiya	Проекция
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Space diagonal	Fazoviy diagonal	Пространственная диагональ
Ambiguous case	Ikki yechimli hol	Неоднозначный случай

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Two sides and the included angle calls for:

- the sine rule
- the cosine rule
- Pythagoras
- the area formula

Three sides given, find an angle — use:

- the sine rule
- the cosine rule rearranged
- Pythagoras
- a protractor

The angle between a line and a plane is measured to:

- any line in the plane
- its projection on the plane
- the normal
- the edge

The space diagonal of an $a \times b \times c$ cuboid is:

- $a + b + c$
- $\sqrt{a^2 + b^2 + c^2}$
- $\sqrt{a^2 + b^2}$
- abc

For a dihedral angle both lines must be perpendicular to:

- each other
- the line of intersection
- the base
- the normal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Space diagonal of a $3 \times 4 \times 12$ box

ANSWER 13

2. Space diagonal of a cube of side 5

ANSWER $5\sqrt{3} \approx 8.66$

3. Base diagonal of a 6×8 rectangle

ANSWER 10

4. Triangle with $a = 5$, $b = 7$, $C = 60°$. Find c

ANSWER $\sqrt{39} \approx 6.24$

5. Area with $a = 6$, $b = 8$, $C = 30°$

ANSWER 12

6. Sine rule: $A = 40°$, $B = 60°$, $a = 10$. Find b

ANSWER ≈ 13.5

7. Angle of a cube's space diagonal with the base

ANSWER $\approx 35.26°$

Medium · the standard question

7 PROBLEMS

1. Cuboid $4 \times 4 \times 7$. Angle of the space diagonal with the base

   ANSWER $\approx 51.1°$

2. Square pyramid, base 6, height 4. Angle of a lateral face with the base

   ANSWER $\approx 53.1°$

3. Same pyramid: angle of a lateral edge with the base

   ANSWER $\approx 43.3°$

4. Triangle $a = 8, b = 5, c = 7$. Find the largest angle

   ANSWER $\approx 81.8°$

5. Triangle $A = 35°, a = 9, b = 12$. Find B

   ANSWER $\approx 49.9°$ or $130.1°$

6. Area of a triangle with sides 6, 7 and included angle $75°$

   ANSWER ≈ 20.3

7. Cube side 4: angle between a face diagonal and a space diagonal

   ANSWER $\approx 35.26°$

Hard · stretch and reason

7 PROBLEMS

1. Cuboid $5 \times 12 \times 9$. Angle between the space diagonal and the 12 edge

   ANSWER $\approx 44.1°$

2. Square pyramid base 10, lateral edge 13. Find the height

   ANSWER $\sqrt{119} \approx 10.9$

3. Same pyramid: dihedral angle between two adjacent faces

   ANSWER $\approx 84.8°$

4. A mast stands at a corner of a 20×30 field; from the opposite corner its angle of elevation is $12°$. Find its height

   ANSWER $\approx 7.66\ m$

5. Regular tetrahedron edge 6: angle between a face and the base

   ANSWER $\approx 70.5°$

6. Regular tetrahedron edge 6: angle between two edges at a vertex

   ANSWER $60°$

7. Cube: prove the space diagonal makes $arctan\ \frac{1}{\sqrt{2}}$ with the base

   ANSWER $tan\ \theta = \frac{a}{a\sqrt{2}}$

07 Homework

Homework — 6 tasks

Draw the extracted triangle separately for every 3-D question.

1. Cuboid $8 \times 6 \times 10$. Find the space diagonal and its angle with the base.
2. Square pyramid, base 12, height 8. Find the slant height and the angle between a lateral face and the base.
3. Same pyramid: find the angle between a lateral edge and the base.

4. Triangle with $a = 11$, $b = 9$, $C = 48°$. Find c and the area.
5. Triangle with sides 7, 8, 12. Find all three angles.
6. Explain, in three sentences, how to find the angle between two planes.

Control work, and the quarter review

The axioms, the positions, the polyhedra, the sections and the trigonometry — all in one paper, then the map that ties them together.

LESSON 17–18 | 2 × 40 MIN | GEOMETRY 10, NAZORAT ISHI 1 | IGCSE E4 REVIEW

LESSON CLOCK

3 | 36' | 11' | 18' | 9' | 3

Instructions	3 min
The paper	36 min
Answers and self-marking	11 min
Rewrite and classify	18 min
Concept map	9 min
Homework	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Present a construction with the rule quoted at each step.
- Classify each lost mark and rewrite the solution.
- Build a concept map of solid geometry so far.

TEXTBOOK

National	pp. 79–82
Cambridge	Core & Extended, pp. 313–316
Workbook	Control paper G1

01 Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	State the three axioms of space and say what each forbids	3	L4–5
2	In a cube, list all edges skew to CD	3	L6–7
3	A polyhedron has $F = 12$ and $E = 30$. Find V, and name it if regular	3	L8–9
4	Construct the section of a cube through A, C and B_1; name the shape and find its area for side 6	6	L12–13
5	Cuboid $6 \times 8 \times 10$: space diagonal and its angle with the base	5	L14–16
6	Square pyramid base 10, height 12: the angle between a lateral face and the base	5	L14–16
7	Count the planes of symmetry of a cube and of a $3 \times 3 \times 7$ cuboid	5	L3

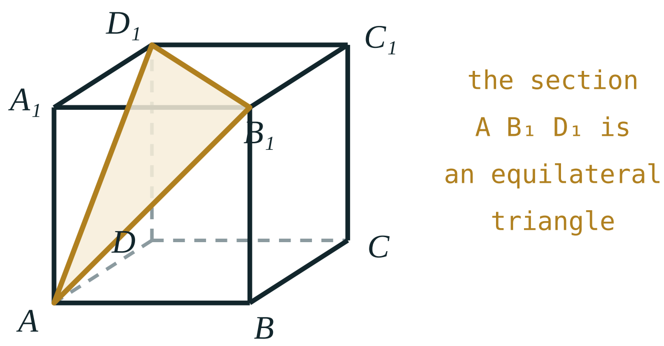
CONSTRUCTIONS ARE MARKED ON REASONS

Q4 gives 2 marks for the correct figure and 4 for the justification. A section drawn without R1/R2/R3 quoted scores a third of the question.

The concept map

Six boxes, and the links between them written as sentences:

- **axioms** → **positions of lines and planes** — “S3 is why two planes meet in a line”
- **positions** → **skew lines** — “the case the plane does not have”
- **polyhedra** → **Euler** — “$F + V - E = 2$ for every convex solid”
- **polyhedra** → **nets** → **surface area**
- **sections** → **R1, R2, R3** — “parallel faces give parallel cuts”
- **3-D trigonometry** → **extract a triangle** — “every 3-D question is a plane question”



One cube carries four of the six boxes: positions, sections, symmetry and 3-D trigonometry.

Looking forward

Quarter II is parallelism of lines and planes — every proof will quote the three axioms of Lesson 4–5 by name, and every construction will use the section rules of Lesson 12–13. Neither set is ever re-taught; they are simply used.

ONE HABIT TO CARRY FORWARD

Draw the solid large, mark the right angles first, then extract the triangle. Almost every mark lost in Q5 and Q6 was lost by working from a diagram three centimetres wide.

02 Worked examples

EXAMPLE 1 Model answer, Q4: section of a cube through A, C, B_1, side 6.

1. AC in the base face
   R1
2. AB_1 in face ABB_1A_1, CB_1 in face BCC_1B_1
   R1 twice
3. All three are face diagonals, $6\sqrt{2}$
   Equilateral triangle
4. $S = \frac{\sqrt{3}}{4} \times 72 = 18\sqrt{3}$

ANSWER

Equilateral triangle, area $18\sqrt{3} \approx 31.2$

EXAMPLE 2 Model answer, Q6: square pyramid base 10, height 12.

1. Apothem of the base 5.
2. Slant height $\sqrt{144 + 25} = 13$.
3. $\cos\theta = \frac{5}{13}$
   Both lines ⊥ to the base edge.
4. $\theta \approx 67.4°$

ANSWER

$\approx 67.4°$

EXAMPLE 3 **Model answer, Q3: $F = 12, E = 30$.**

1. $V = 2 - F + E = 2 - 12 + 30$
2. $V = 20$
3. 12 faces, 20 vertices, 30 edges.

   The dodecahedron.

ANSWER

$V = 20$ — the regular dodecahedron

03 Interactive model

Work Q4 and Q6 on the board before the rewrite hour begins.

The quarter in ten questions

Three non-collinear points determine:

- no plane
- one plane
- two
- infinitely many

Edges of a cube skew to a given edge:

- 2
- 3
- 4
- 6

$F = 12, E = 30$ gives V:

- 12
- 16
- 20
- 24

Section through A, C, B_1 of a cube:

- square
- rectangle
- equilateral triangle
- hexagon

Space diagonal of a $6 \times 8 \times 10$ cuboid:

- $10\sqrt{2}$
- 24
- 14
- 20

Planes of symmetry of a cube:

- 3
- 6
- 9
- 12

Planes of symmetry of a $3 \times 3 \times 7$ cuboid:

- 3
- 4
- 5
- 9

A plane cuts two parallel faces in:

- equal lines
- parallel lines
- perpendicular lines
- points

The maximum sides of a cube section:

- 4
- 5
- 6
- 8

The angle between a line and a plane uses:

- the normal
- the projection
- any line in the plane
- the edge

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Assessment	Baholash	Оценивание
Justification	Asoslash	Обоснование
Construction step	Yasash bosqichi	Шаг построения
Concept map	Tushunchalar xaritasi	Карта понятий
Revision	Takrorlash	Повторение
Target	Maqsad	Цель
Self-assessment	Oʻz-oʻzini baholash	Самооценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A construction scores most of its marks for:

- neatness
- the justification
- colour
- speed

Euler’s formula fails for:

- a cube
- a prism
- a solid with a hole
- a pyramid

Before any 3-D trigonometry you should:

- guess
- extract and redraw the triangle
- measure
- use a calculator

Skew lines differ from parallel lines because:

- they meet
- no plane contains both
- they are equal
- they are perpendicular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. State axiom S1

   ANSWER Three non-collinear points determine exactly one plane

2. Edges skew to CD in a cube

   ANSWER $AA_1, BB_1, A_1D_1, B_1C_1$

3. $F = 12, E = 30$; find V

   ANSWER 20

4. Space diagonal of a $6 \times 8 \times 10$ cuboid

   ANSWER $10\sqrt{2}$

5. Planes of symmetry of a cube

   ANSWER 9

6. Section through A, C, B_1

   ANSWER equilateral triangle

7. Slant height of a pyramid, base 10, height 12

   ANSWER 13

Medium · the standard question

7 PROBLEMS

1. Area of the section through A, C, B_1 for side 6

   ANSWER $18\sqrt{3}$

2. Angle of a $6 \times 8 \times 10$ space diagonal with the base

   ANSWER $45°$

3. Angle between a lateral face and the base, pyramid 10/12

   ANSWER $\approx 67.4°$

4. Planes of symmetry of a $3 \times 3 \times 7$ cuboid

   ANSWER 5

5. F, V, E of an octagonal prism

   ANSWER $10, 16, 24$

6. Diagonal section area, cube side 6

   ANSWER $36\sqrt{2}$

7. Angle of a lateral edge with the base, pyramid base 10 height 12

   ANSWER $\approx 59.5°$

Hard · stretch and reason

7 PROBLEMS

1. Cube side 6: area of the midpoint hexagonal section

ANSWER $27\sqrt{3} \approx 46.8$

2. Prove no cube section is a heptagon

ANSWER Six faces give at most six sides

3. A solid has 32 faces, 12 pentagons and 20 hexagons, 3 at each vertex. Find V, E

ANSWER $V = 60, E = 90$

4. Cuboid $a \times a \times b$: find the angle between the space diagonal and a base edge

ANSWER $\arccos \frac{a}{\sqrt{2a^2 + b^2}}$

5. Regular tetrahedron edge a: find the angle between two faces

ANSWER $\approx 70.53°$

6. Construct the section of a cube through the midpoints of AB, BC and CC_1

ANSWER A pentagon

7. Explain why a plane cannot meet a face of a convex solid in two separate segments

ANSWER Convexity makes the intersection connected

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite in full every control-work question that lost a mark, with reasons.
2. Finish the concept map with all six links written as sentences.
3. Cuboid $5 \times 12 \times 9$: find the space diagonal, its angle with the base and its angle with the longest edge.

4. Write your target for Quarter II in one checkable sentence.